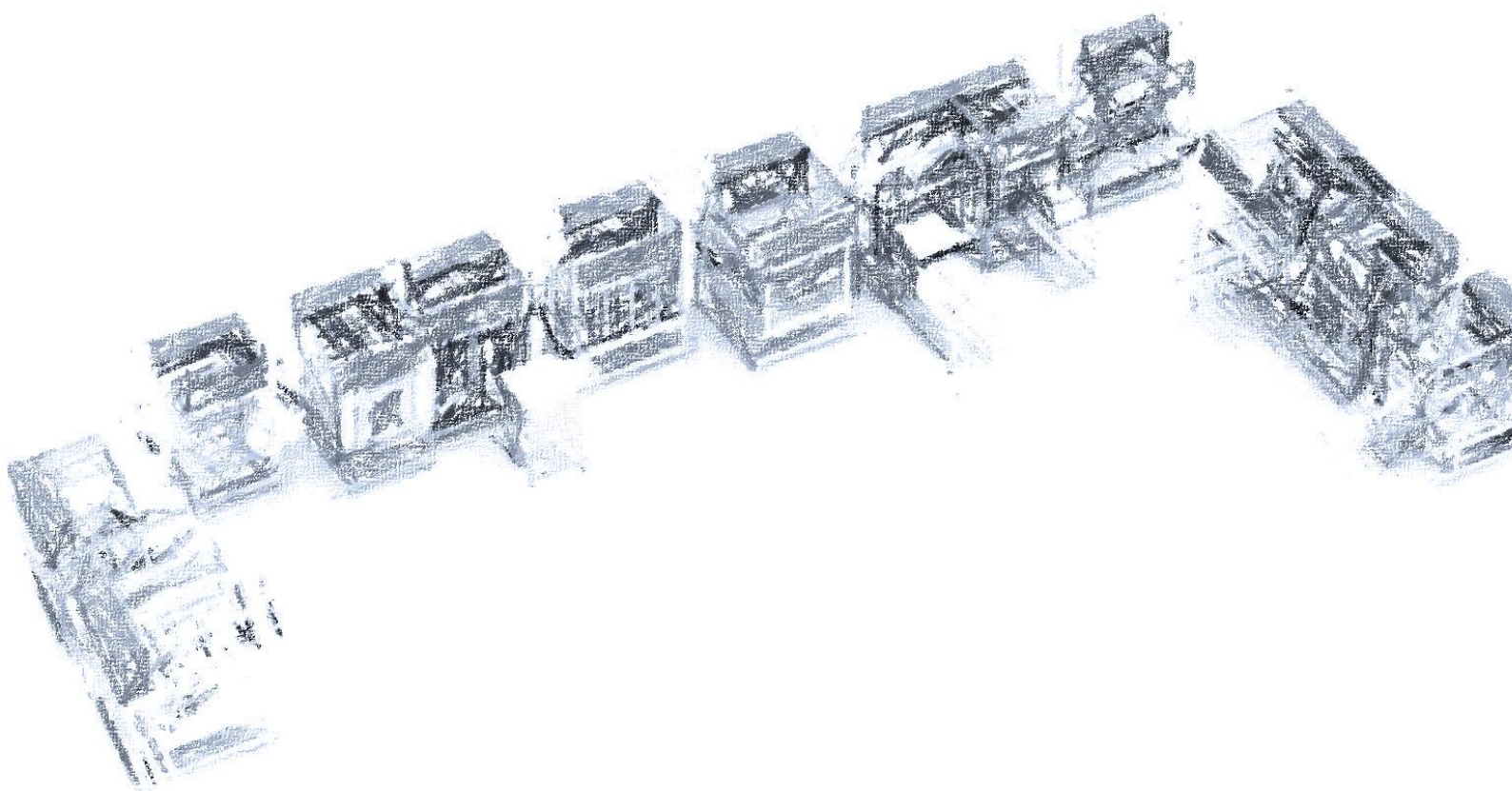


## **BUDGET QUOTE C610222 - 1**



## **MODULAR ALL DRIVES TEST SYSTEM**

To:

**WEG DRIVES & CONTROLS -  
AUTOMACAO LTDA**

AV. PREFEITO WALDEMAR GRUBBA,  
3000-VILA LALAU,  
89256900 - JARAGUA' DO SUL -  
BRAZIL

**FAO** Aluizio Augusto Kleine Kirschner

SUBJECT: **MODULAR ALL DRIVES TEST SYSTEM**

Our **BUDGET QUOTE: C610222 - 1**

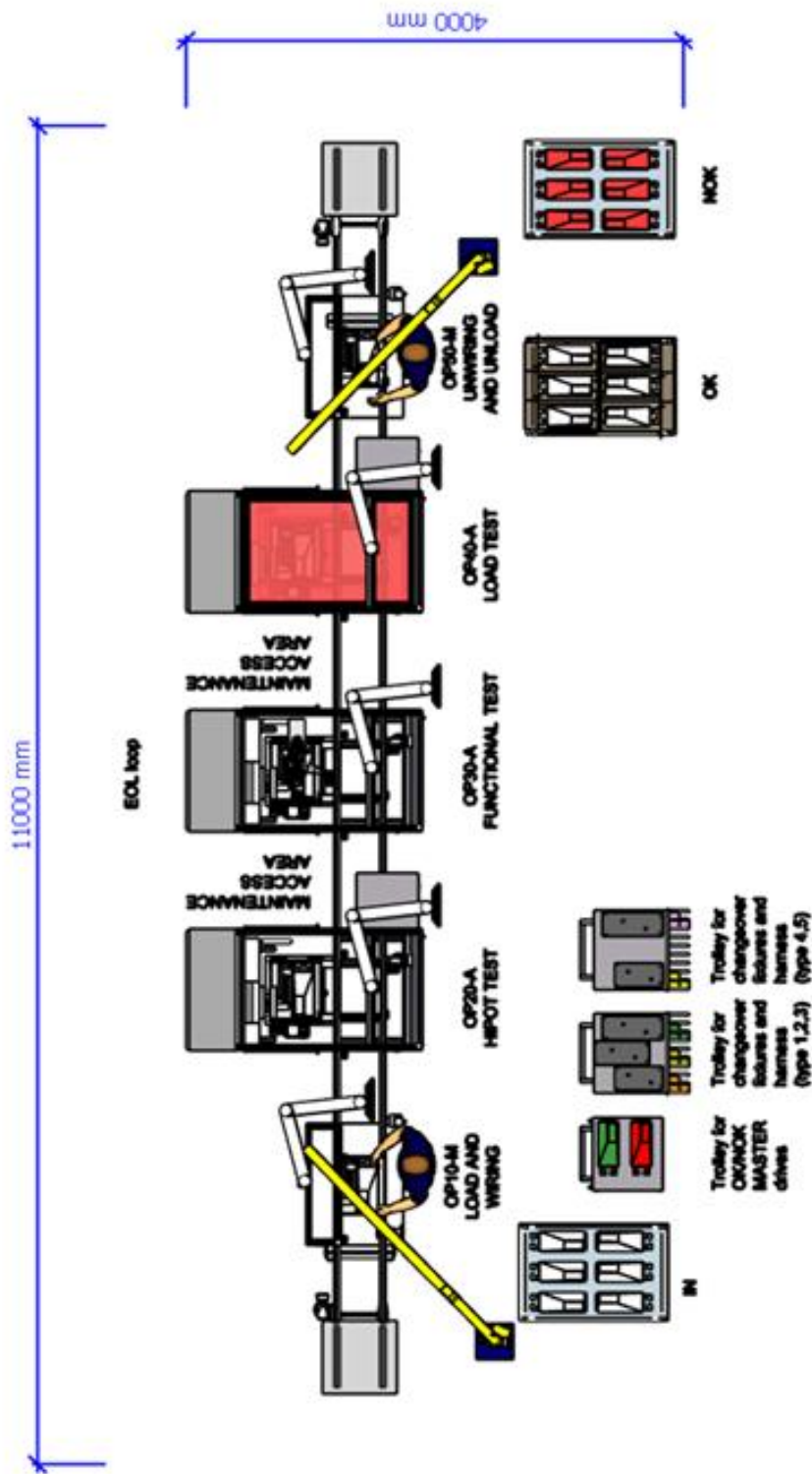
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With reference to your kind request for quote (dated 2026/06/10), your technical specification there attached (rev.00, dated May 2026) and technical meeting dated 2026/06/16, here follows our **BUDGET** quote for the study and supply of the items in subject.

## **SCOPE**

Scope of this proposal is the supply of n.1 test line for drives, to be installed at Your plant of Santa Catarina (Brazil) according to details below.

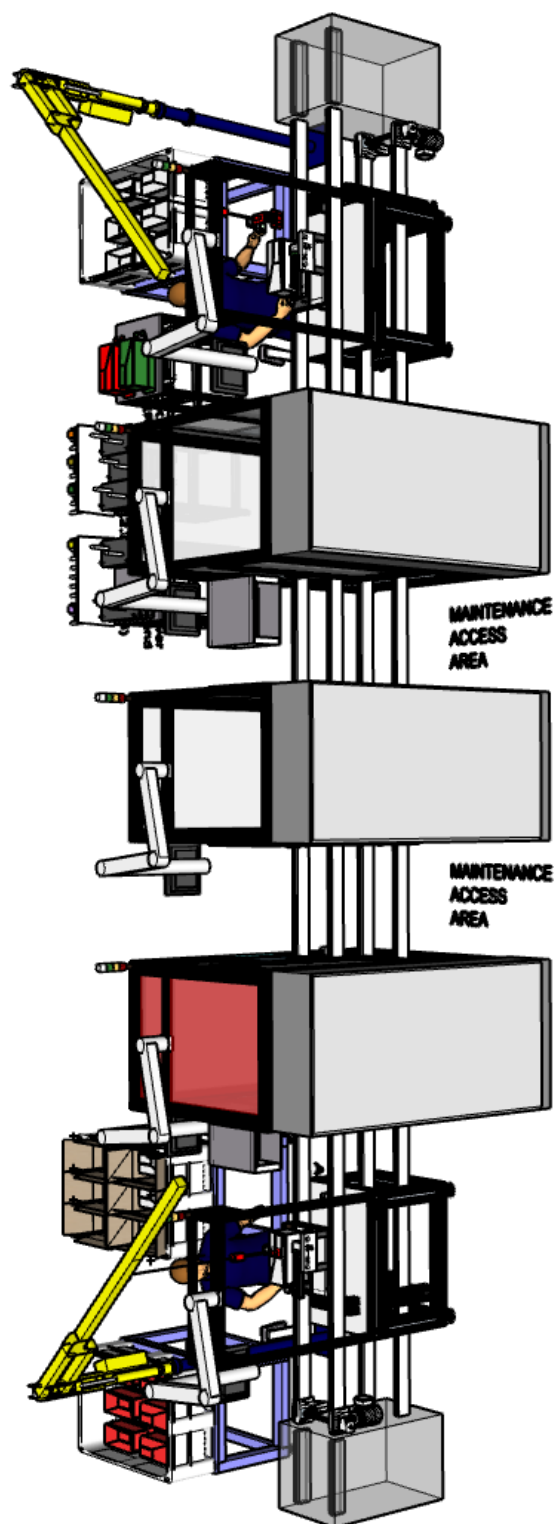
M.1 - LAYOUT DRAFT



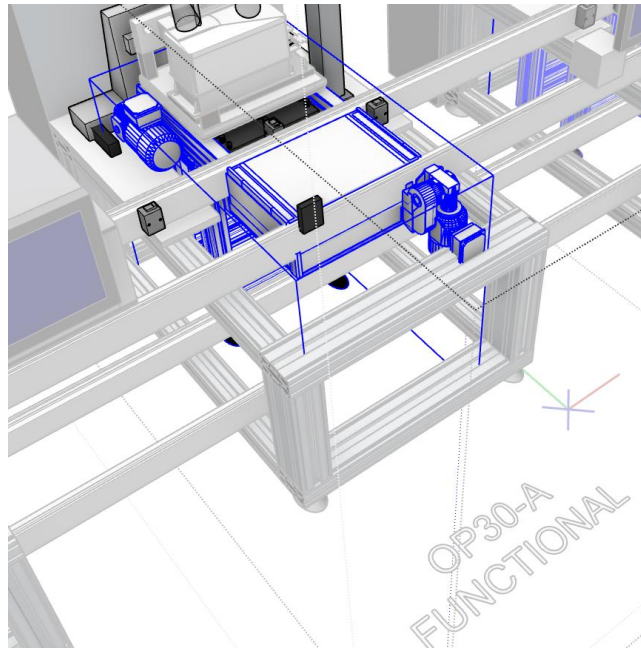
2D top view







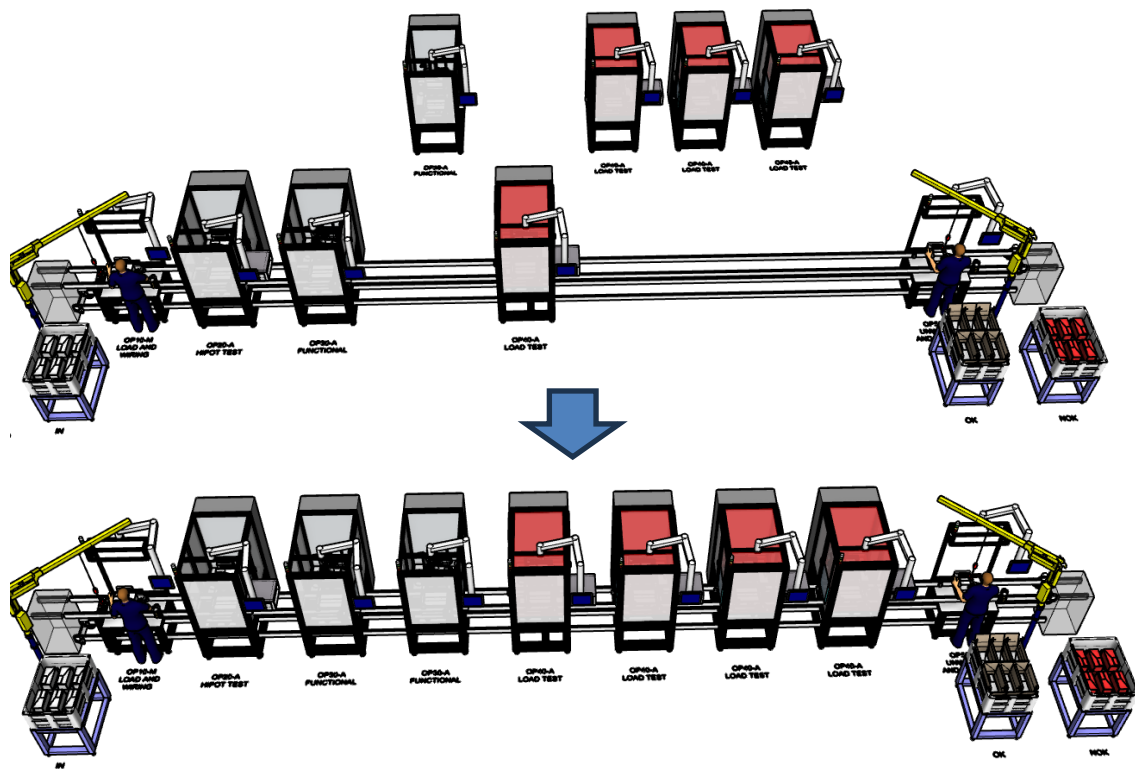




*Focus on conveyor transversal "add-on" architecture with respect to the existing main conveyor flow.*

## LAYOUT FLEXIBILITY

By means of above showed architecture, and by proper initial definition of conveyor length (eventually greater than length currently included in base proposal), additional test modules can be added in a following stage.





**COMPOSITION OVERVIEW**

Our supply will consist of the following equipment.

**M.1 - Base proposal:**

|              |                                                                                  |
|--------------|----------------------------------------------------------------------------------|
| <b>OP00</b>  | Conveyor system; 3 pallets and relevant changeover equipment; Masters management |
| <b>OP20A</b> | HIPOT test module                                                                |
| <b>OP30A</b> | Functional test module                                                           |
| <b>OP40A</b> | Load test module                                                                 |
| <b>S</b>     | Server                                                                           |

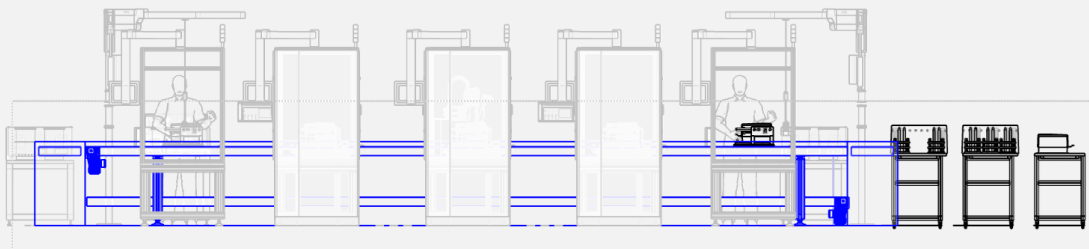
**Options:**

|               |                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------|
| <b>Opt.1</b>  | Manual wiring and load/unload                                                                                    |
| <b>Opt.2A</b> | 1 week on-site support (1 technician x 5 working days on central shift, travel and boarding expenses included)   |
| <b>Opt.2B</b> | 1 week extension of on-site support (1 technician x 5 working days on central shift, boarding expenses included) |
| <b>Opt.3A</b> | Additional pallet                                                                                                |
| <b>Opt.3B</b> | Changeover equipment: for 1 typology, for 1 pallet                                                               |

## DETAILED SUPPLY COMPOSITION

**OP00 :** Conveyor system / 3 pallets and relevant changeover equipment / Masters management

### F01 : Conveyor loop



*Focus on conveyor loop*

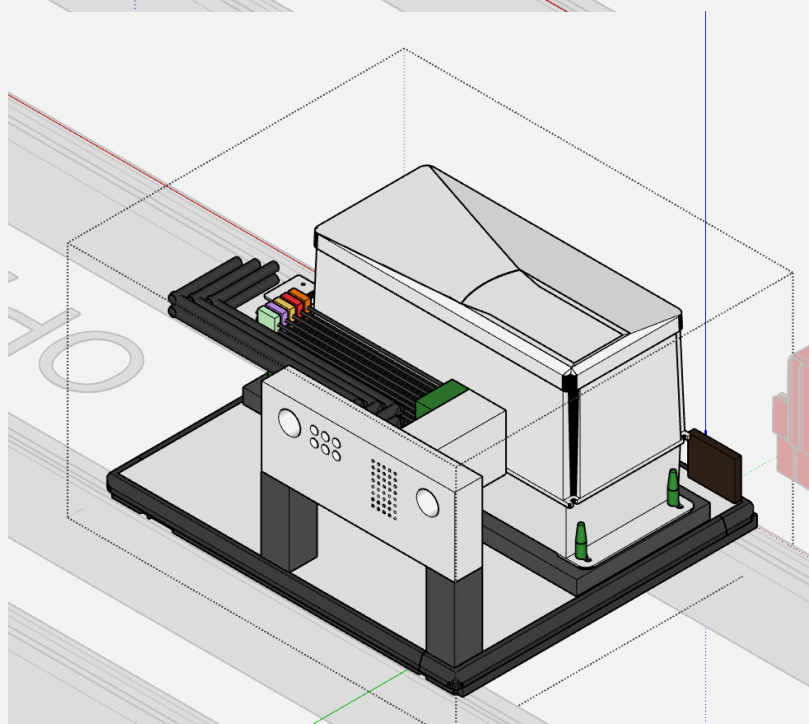
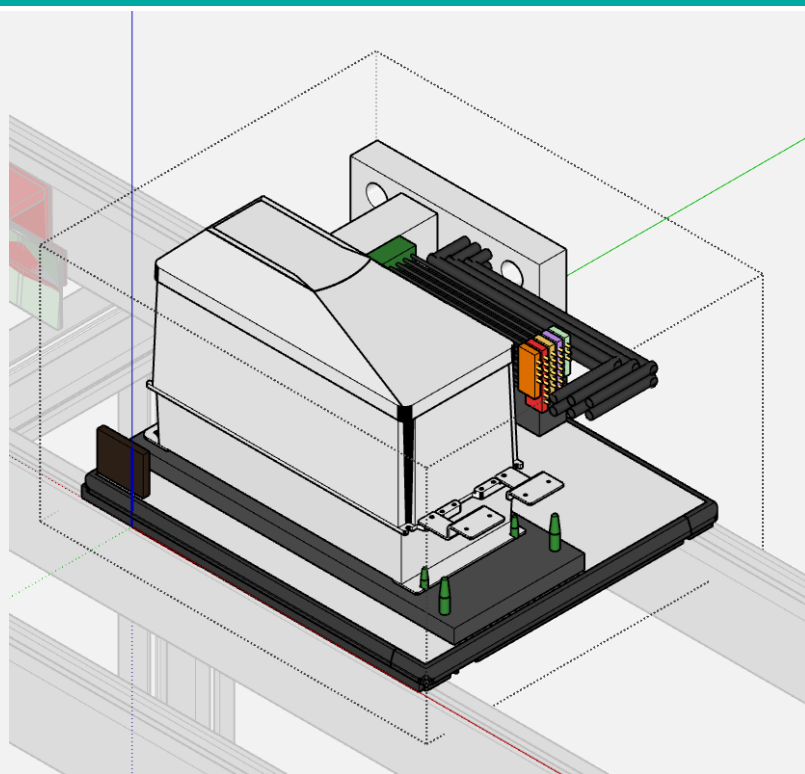
#### Function details

- 1 x BOSCH TS2 (or similar brand / model) Fly-roller chain conveyor system according flow and dimensions showed on layout. It includes 9m upper conveyor ( pprox. 900mm height level), 9m bottom conveyor ( pprox. 300mm height level), 2 electrical lifters for loop closing, 6 pallet stop devices for flow management.
- 1 x Remote I/O, motors management, accessories
- 1 x Pneumatic valves and accessories

### F02 : Prototype pallet

#### Function details

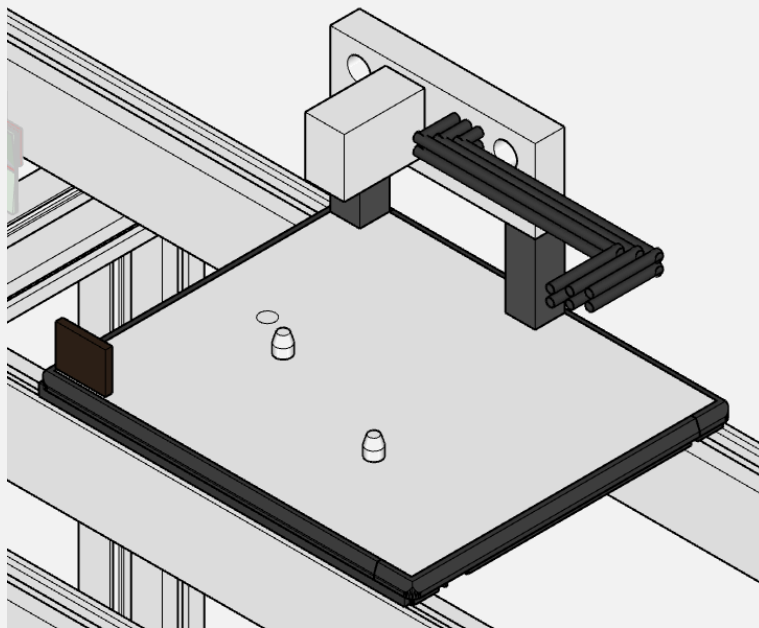
- 1 x Activities related with 1<sup>st</sup> prototype pallet building

**F03 : Series pallet**

*Pallet concept*

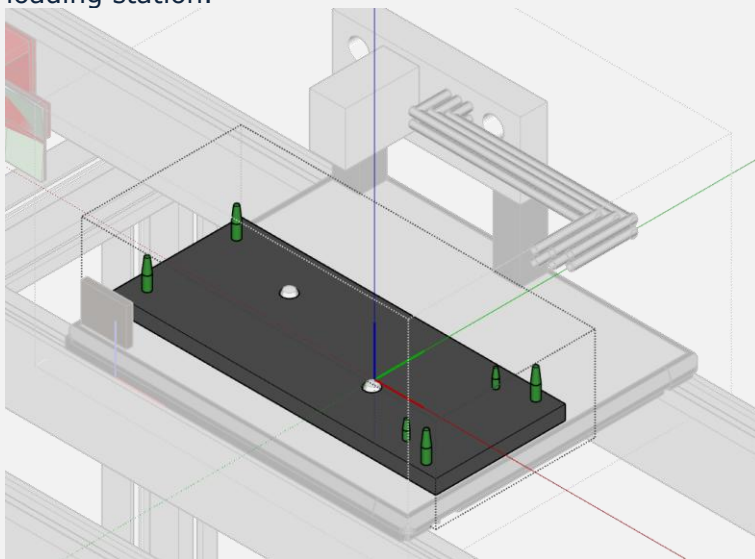
**Function details**

- 3 x Pallet base featured by dimensions  $\leq 400 \text{ mm} \times 480 \text{ mm}$
- 3 x Base pallet machining and centring pins (2x) for changeover fixture nests
- 3 x Match plate (STAUBLI Combitaq or similar brand/model) including 3 x AC IN connections + 3 x AC OUT connections + signal connections (D-I/O, A-I/O, Bus, STO etc) & AC HV harness & LV connector for changeover harness
- 1 x RFID TAG memory



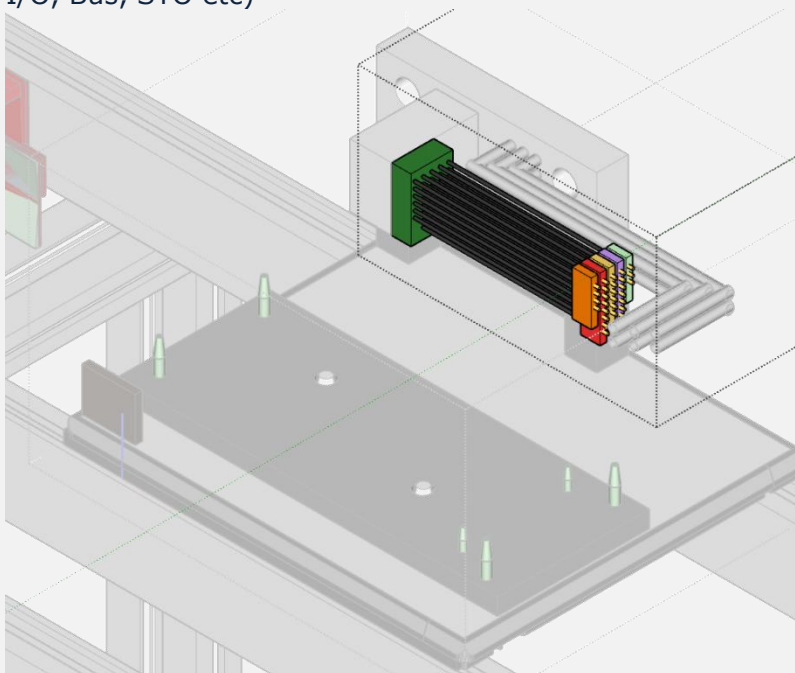
*3D concept of above mentioned details*

- 15 x Changoever fixture nests (5 types x 3 pallets) including centring pins for specific drive model. Fixtures will include 1 QR code for typology check at loading station.



*Changeover fixture*

- 15 x Changeover harness (5 types x 3 pallets) for signal connections (D-I/O, A-I/O, Bus, STO etc)



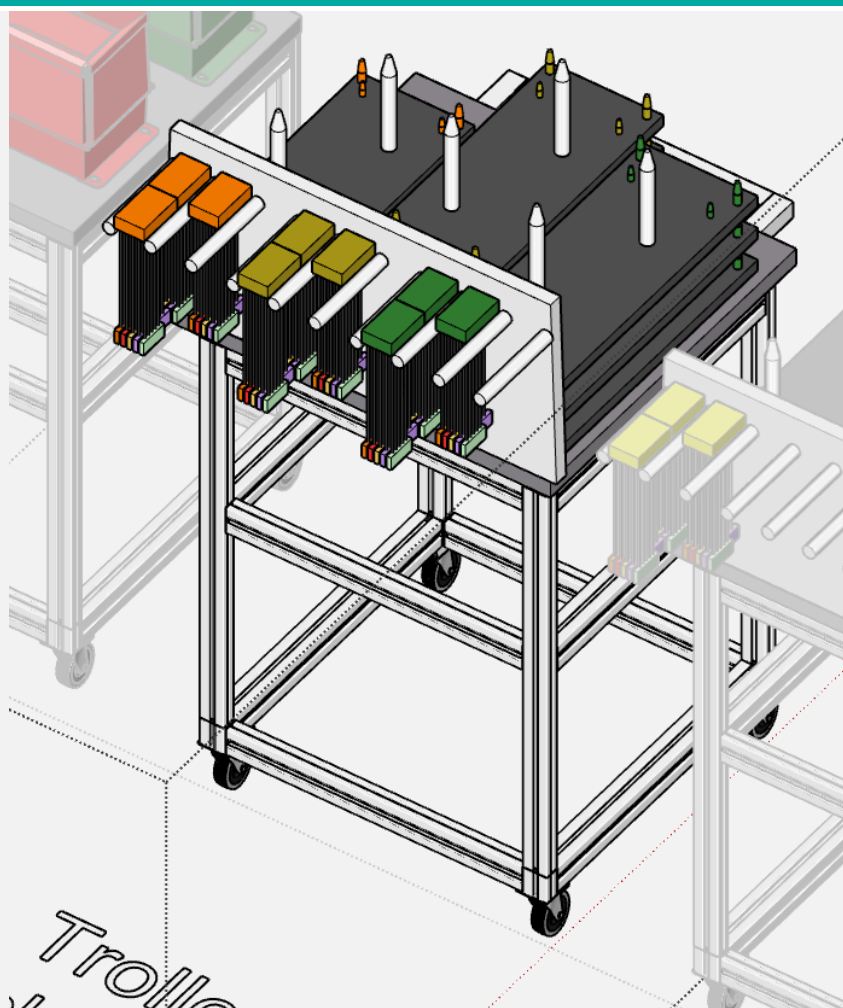
*Changeover signal harness*

**F04 : General electrical equipment**
**Function details**

1 x Electrical wires ducts

**F05 : General pneumatic equipment**
**Function details**

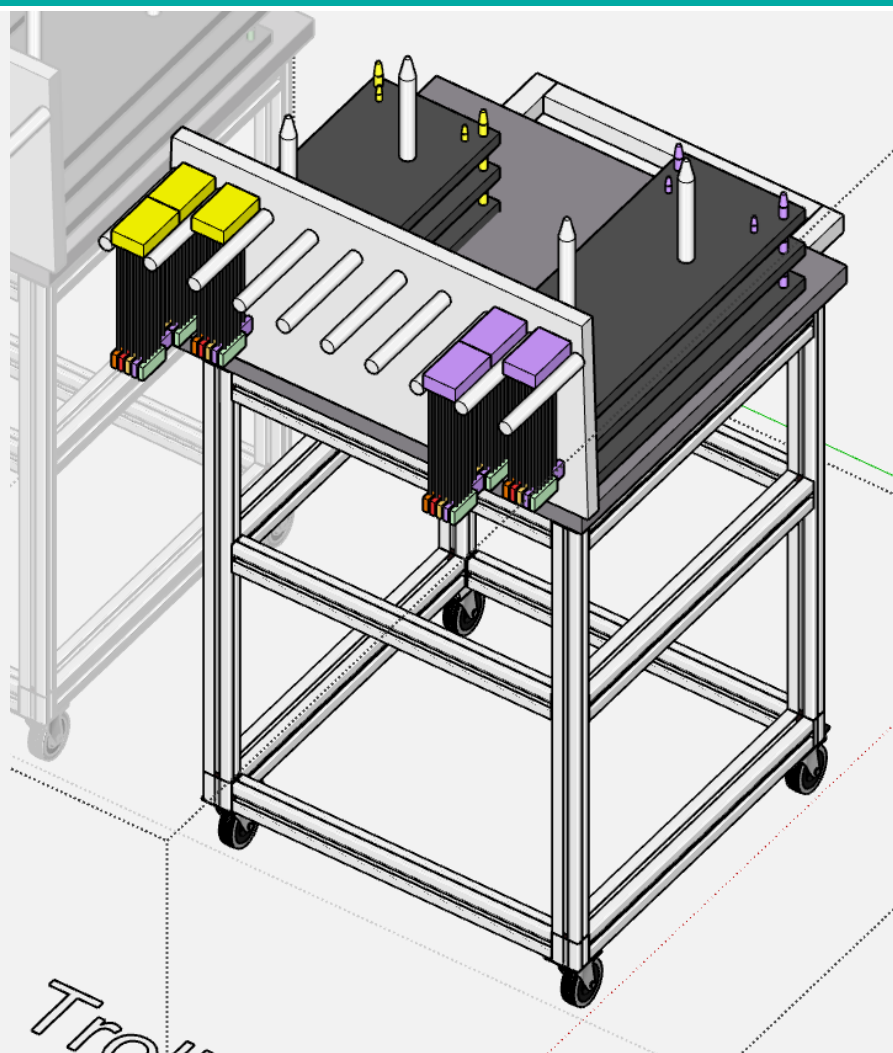
1 x Pneumatic ducts

**F06 : Trolley for pallet changeover fixtures & wiring (type 1,2,3)**

**Function details**

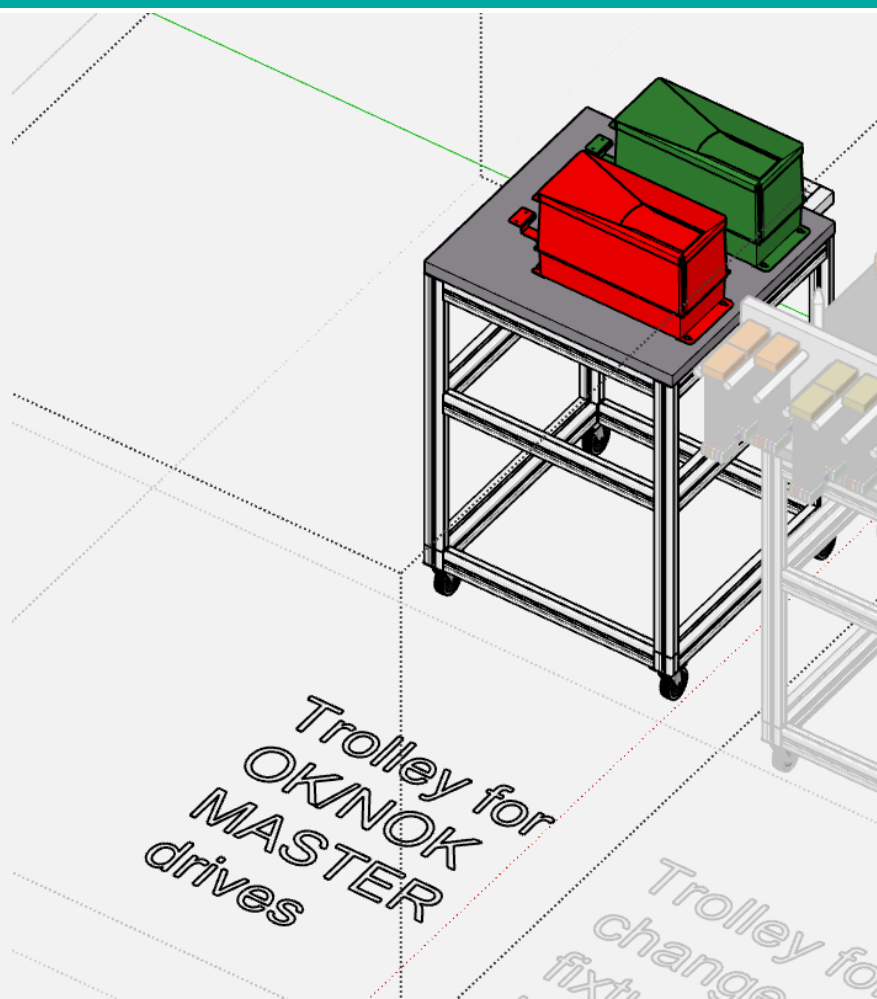
1 x Trolley base plate including supports for n.9 changeover fixture nests and harness (3 x 3 typologies)

1 x Trolley base frame

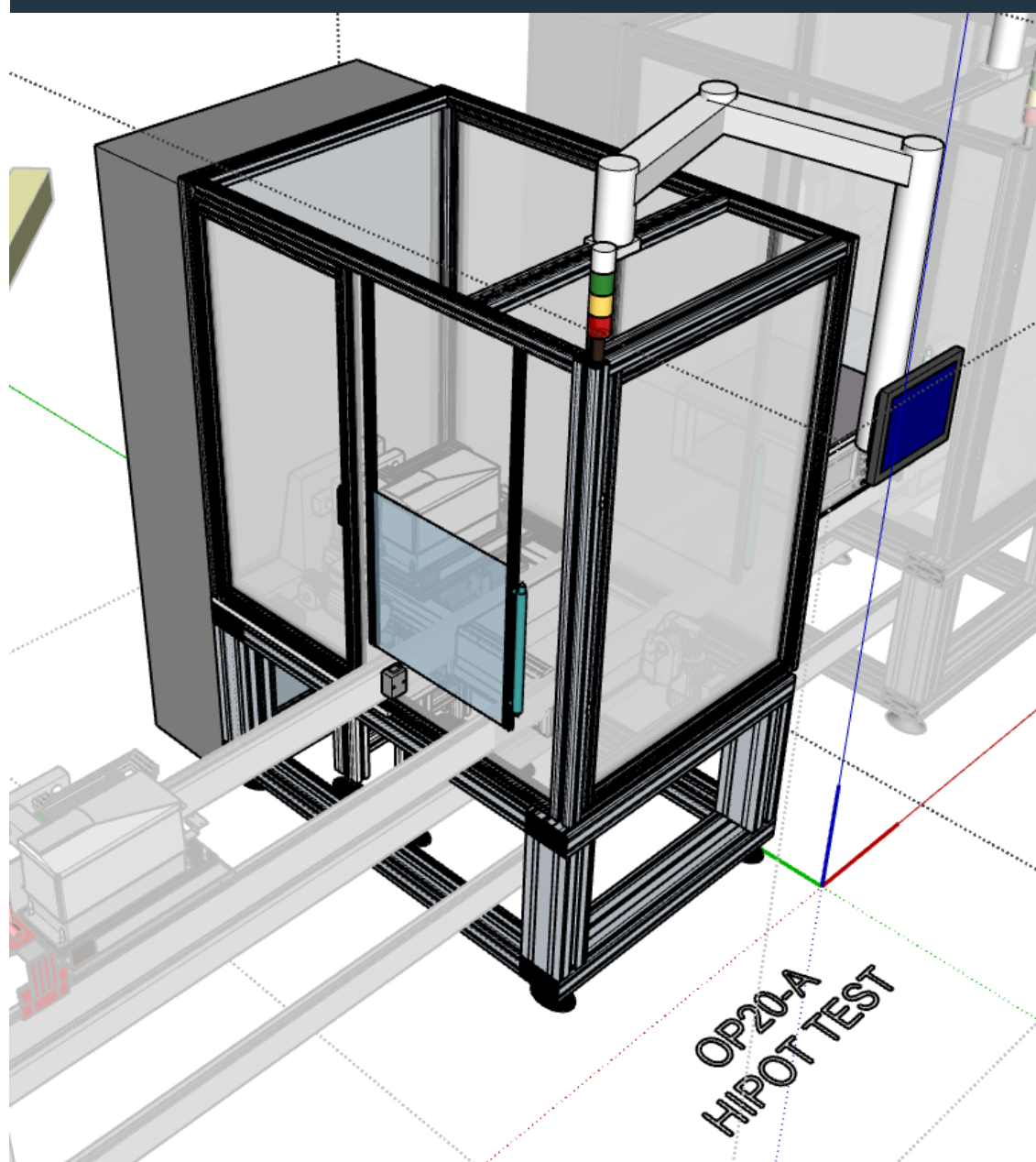


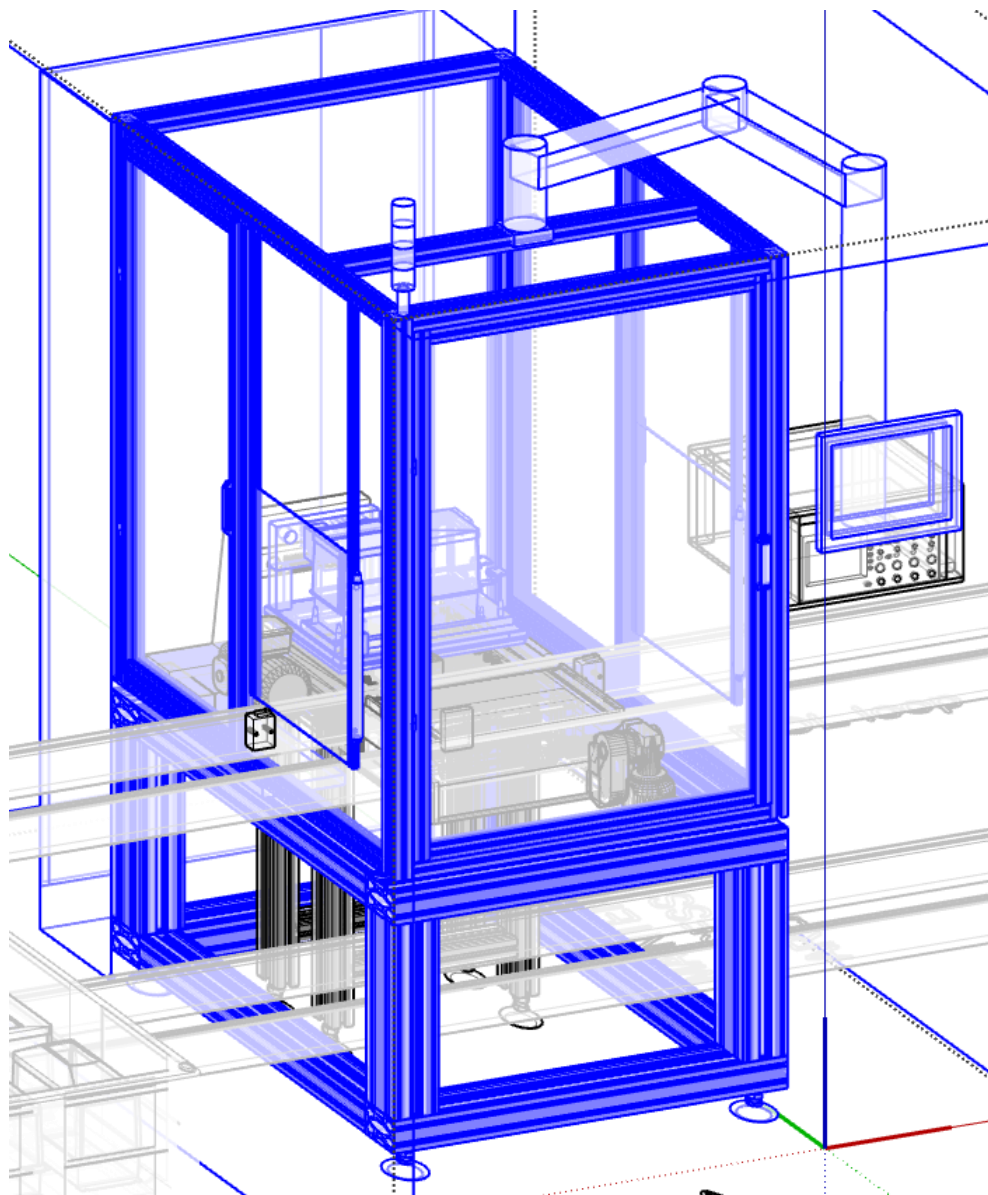
**F07 : Trolley for pallet changeover fixtures & wiring (type 4,5)**

**Function details**

- 1 x Trolley base plate including supports for n.6 changeover fixture nests and harness (3 x 2 typologies)
- 1 x Trolley base frame

**F08 : Masters management: Trolley for master drives****Function details**

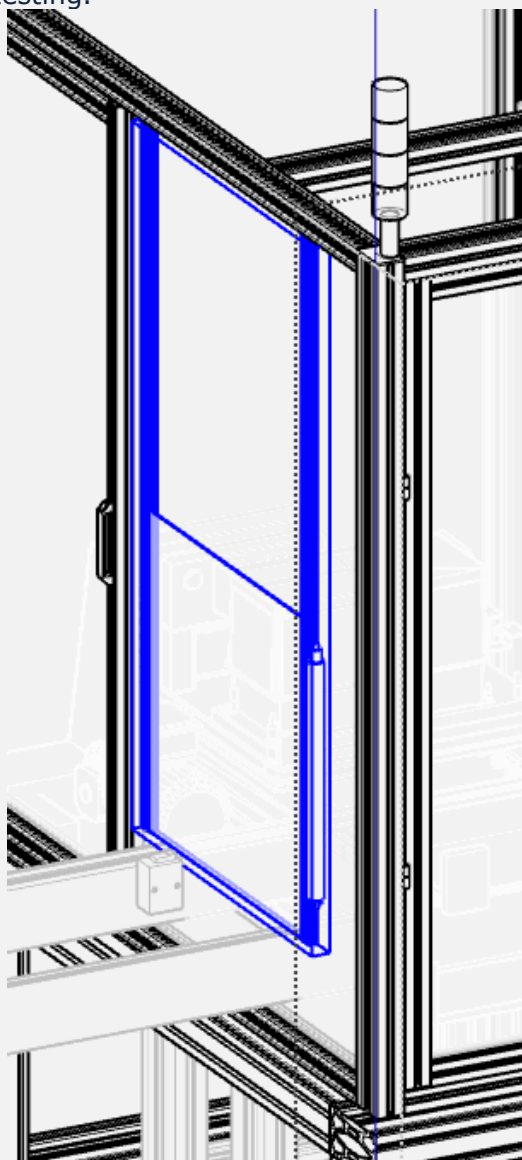
- 1 x Trolley base frame
- 1 x Trolley base plate including supports for n.2 master parts (OK / NOK; master parts considered to be delivered by WEG)

**OP20A : HIPOT test module**

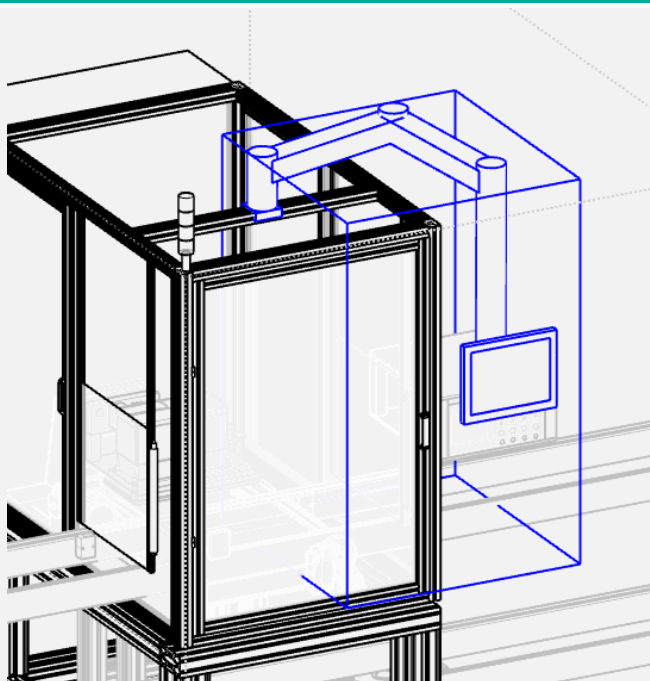
**OP20A : General equipment****F01 : Machine structure****Function details**

- 1 x Aluminium profiles bottom structure featured by dimensions showed on layout
- 1 x Upper structure realized with aluminium profiles and polycarbonate panels.
- 1 x Steel base plate

- 1 x Pneumatic protection panel (left side); open during pallet movement / closed during testing.



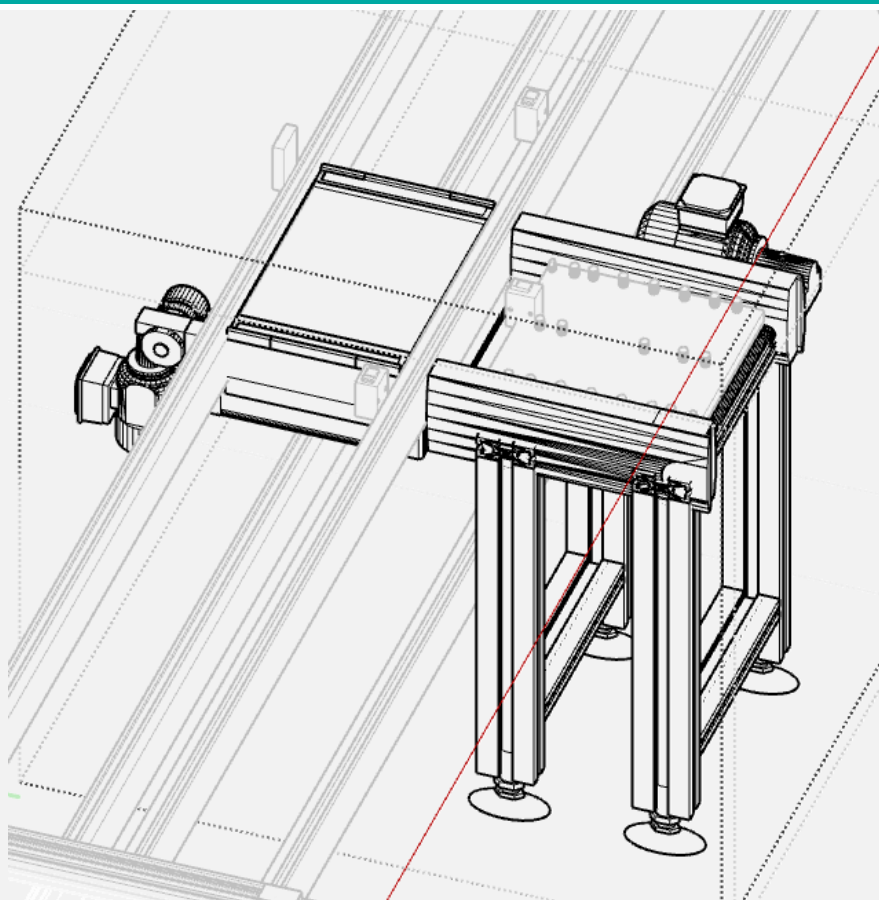
- 1 x Pneumatic protection panel (right side); open during pallet movement / closed during testing.
- 3 x Door interlock
- 1 x Pushbuttons, emergency buttons and other general electrical accessories
- 1 x Machine lighting
- 1 x Pushbuttons, Lightning, Andon light (showing machine status: emergency or failure / automatic mode / piece waiting)
- 1 x Air treatment unit, flow switch, shut-off valve enabling air interception upstream machine.

**F02 : Operator panel****Function details**

- 1 x Supporting arm
- 1 x WEG 15,6" cMTX series HMI

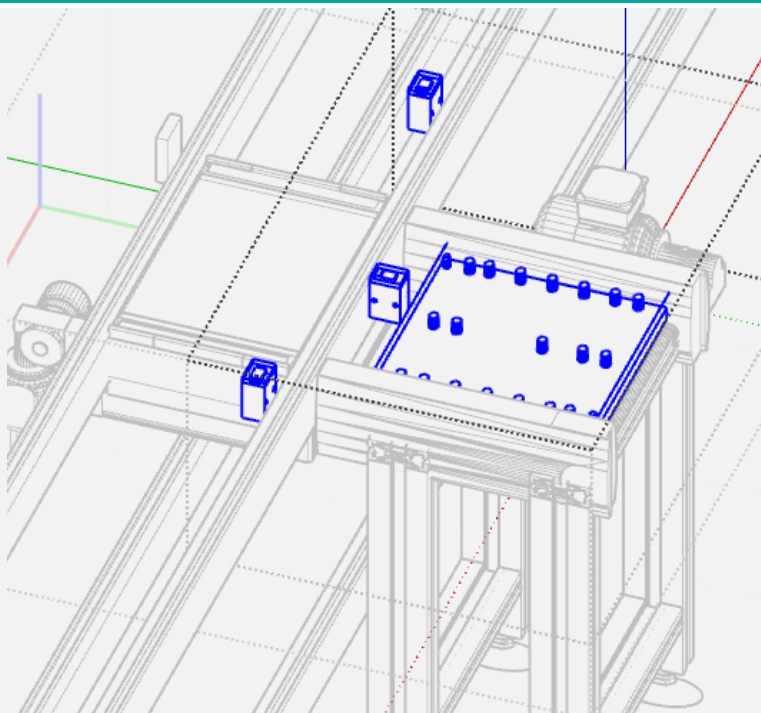
**F03 : Electrical cabinet****Function details**

- 1 x Cabinet, general sectioning, safety and protection components, indicator lights, user sockets. Cabinet power supply will be: 380 Vac 60 Hz power supply (auxiliaries: 24 VDC), PLC WEG. The PLC will perform essentially the pallet IN/OUT and start cycle. The test routine will be fully executed by PC.

**OP20A : Functional equipment****F01 : Pallet conveying into station****Function details**

- 1 x TS2 Bosch Rexroth 90° transverse conveyor unit (or similar brand/model)  
+ 700mm reversible conveyor segment
- 1 x Remote I/O, motors management, accessories
- 1 x Pneumatic valves and accessories



**F02 : Pallet management****Function details**

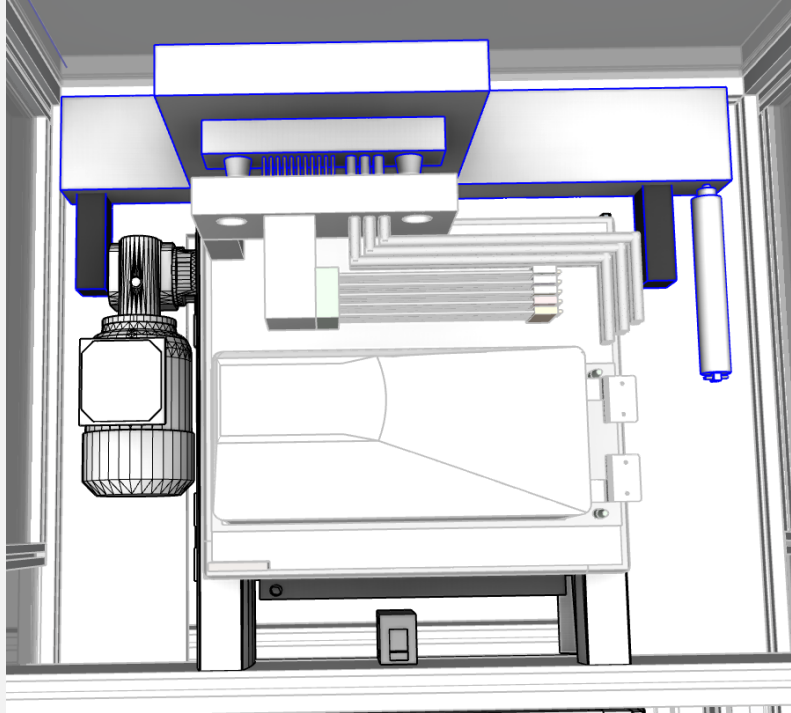
- 3 x Pallet stop pneumatic device
- 1 x Pneumatic index device for pallet in test position
- 1 x Remote I/O
- 1 x Pneumatic valves and accessories

**F03 : RFID code reading****Function details**

- 1 x Supporting frame
- 1 x RFID code reading unit

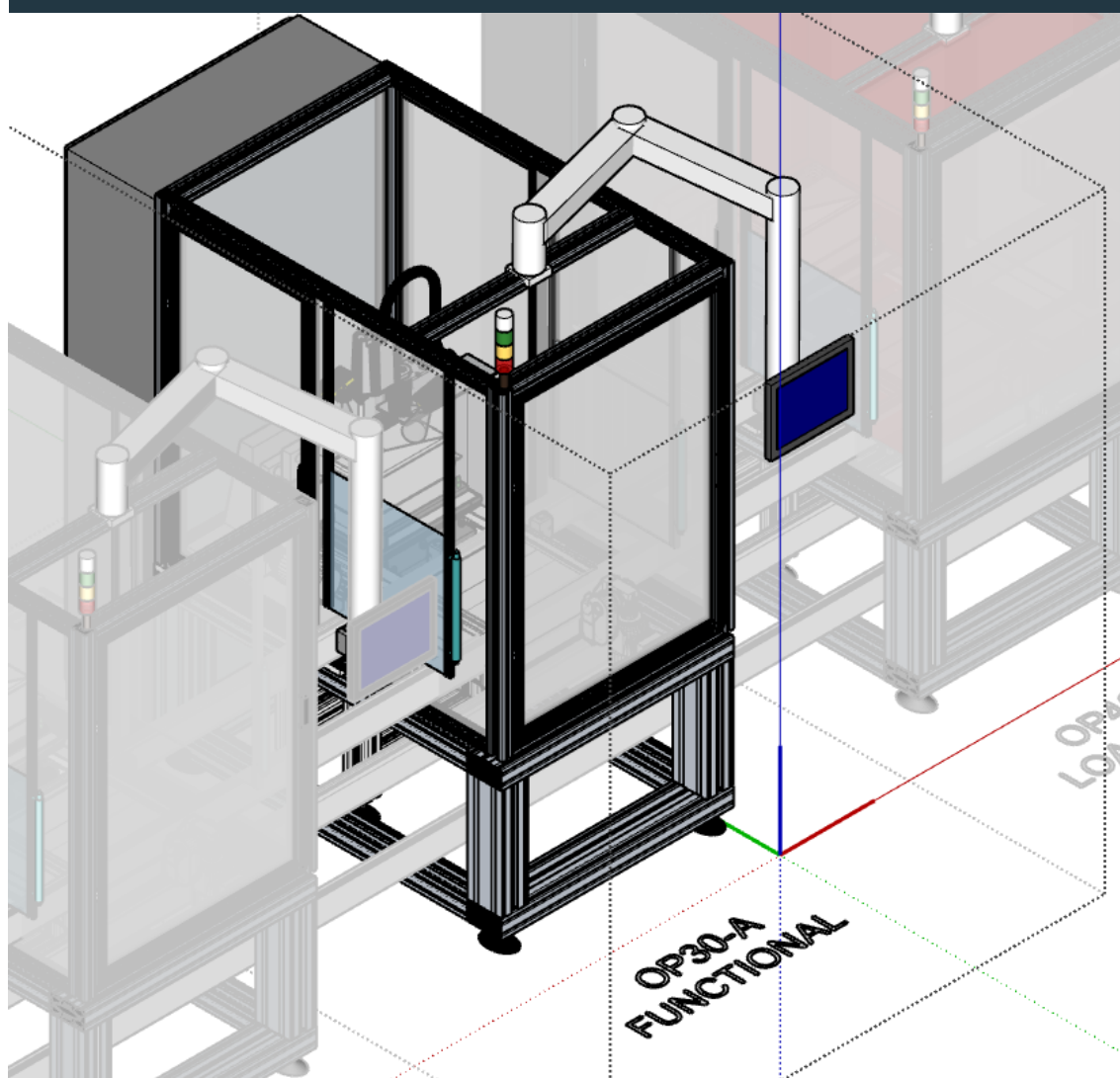
**F04 : Electrical test**
**Function details**

- 1 x Movement system for connection unit (pneumatic)
- 1 x Connection unit towards pallet match plate (STAUBLI Combitaq or similar brand/model).



*Focus on connection unit and relevant movement system*

- 1 x Industrial PC with I/O boards
- 1 x HIPOT test unit & HW channels management: Chroma 19020 Multi-Channel Hipot Tester (or equivalent brand/model). Specialized high-dielectric wiring. *10 channel configuration for complete AC/DC/IR testing; 19" rackmount kit, communication card, complete set of 10 channel shielded high voltage test leads; Safety interlock management, and pass/fail signaling.*  
*This preliminary hardware selection represents our baseline technical proposal for the testing station. The exact scope of supply is to be confirmed or modified (even simplified) upon receipt of the detailed technical specifications.*
- 1 x Hardware matrix for safe residual charge discharge and automatic grounding at the end of the test cycle.
- 1 x Remote I/O
- 1 x Pneumatic management

**OP30A : Functional test module**

**OP30A : General equipment**
**F01 : Machine structure**
**Function details**

- 1 x Aluminium profiles bottom structure featured by dimensions showed on layout
- 1 x Upper structure realized with aluminium profiles and polycarbonate panels.
- 1 x Steel base plate
- 1 x Pneumatic protection panel (left side); open during pallet movement / closed during testing.
- 1 x Pneumatic protection panel (right side); open during pallet movement / closed during testing.
- 3 x Door interlock
- 1 x Pushbuttons, emergency buttons and other general electrical accessories
- 1 x Machine lighting
- 1 x Pushbutons, Lightning, Andon light (showing machine status: emergency or failure / automatic mode / piece waiting)
- 1 x Air treatment unit, flow switch, shut-off valve enabling air interception upstream machine

**F02 : Operator panel**
**Function details**

- 1 x Supporting arm
- 1 x WEG 15,6" cMTX series HMI

**F03 : Electrical cabinet**
**Function details**

- 1 x Cabinet, general sectioning, safety and protection components, indicator lights, user sockets. Cabinet power supply will be: 380 Vac 60 Hz power supply (auxiliaries: 24 VDC), PLC WEG. The PLC will perform essentially the pallet IN/OUT and start cycle. The test routine will be fully executed by PC.

## OP30A : Functional equipment

### F01 : Pallet conveying into station

#### Function details

- 1 x TS2 Bosch Rexroth 90° transverse conveyor unit (or similar brand/model)  
+ 700mm reversible conveyor segment
- 1 x Remote I/O, motors management, accessories
- 1 x Pneumatic valves and accessories

### F02 : Pallet management

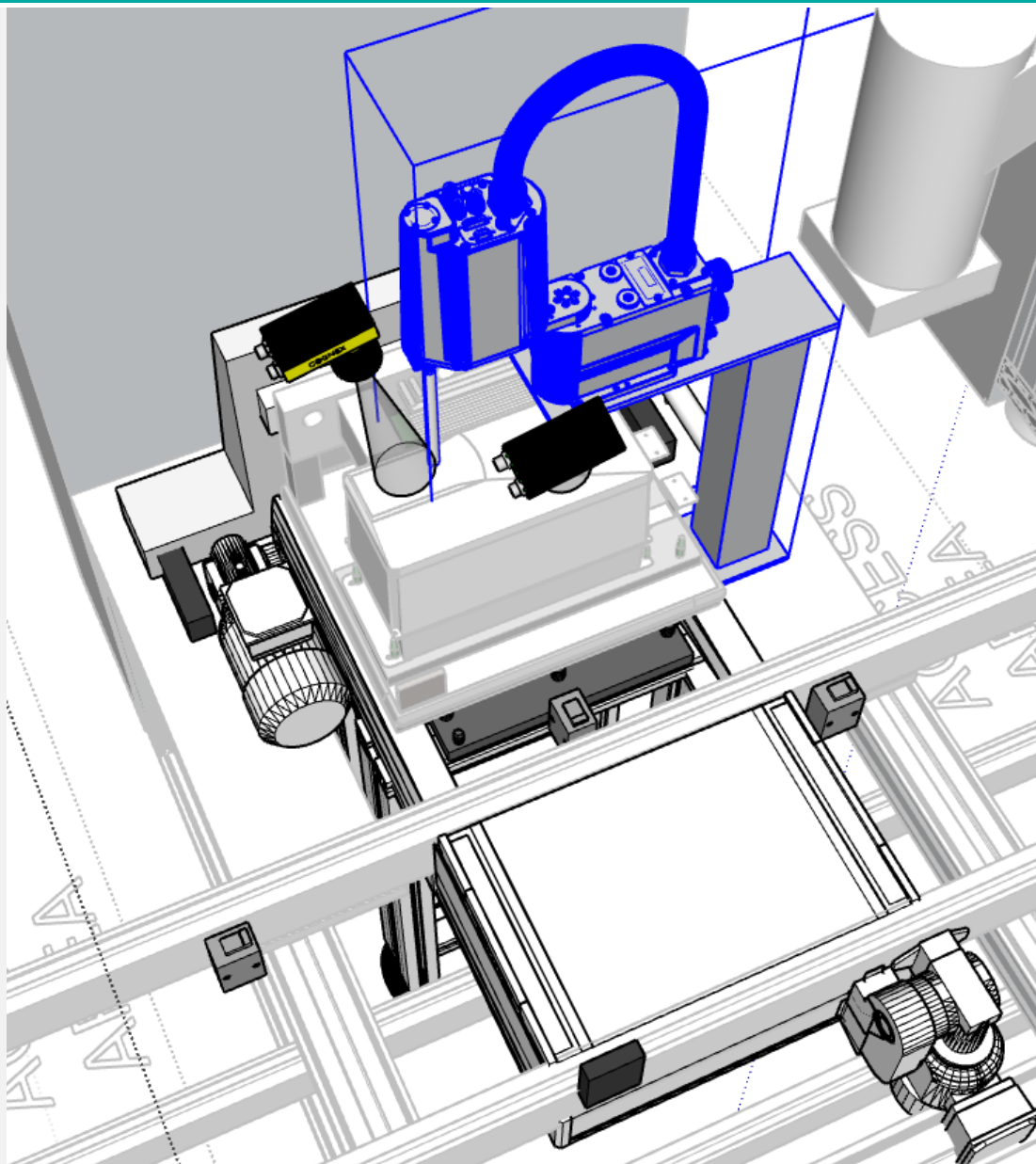
#### Function details

- 3 x Pallet stop pneumatic device
- 1 x Pneumatic index device for pallet in test position
- 1 x Remote I/O
- 1 x Pneumatic valves and accessories

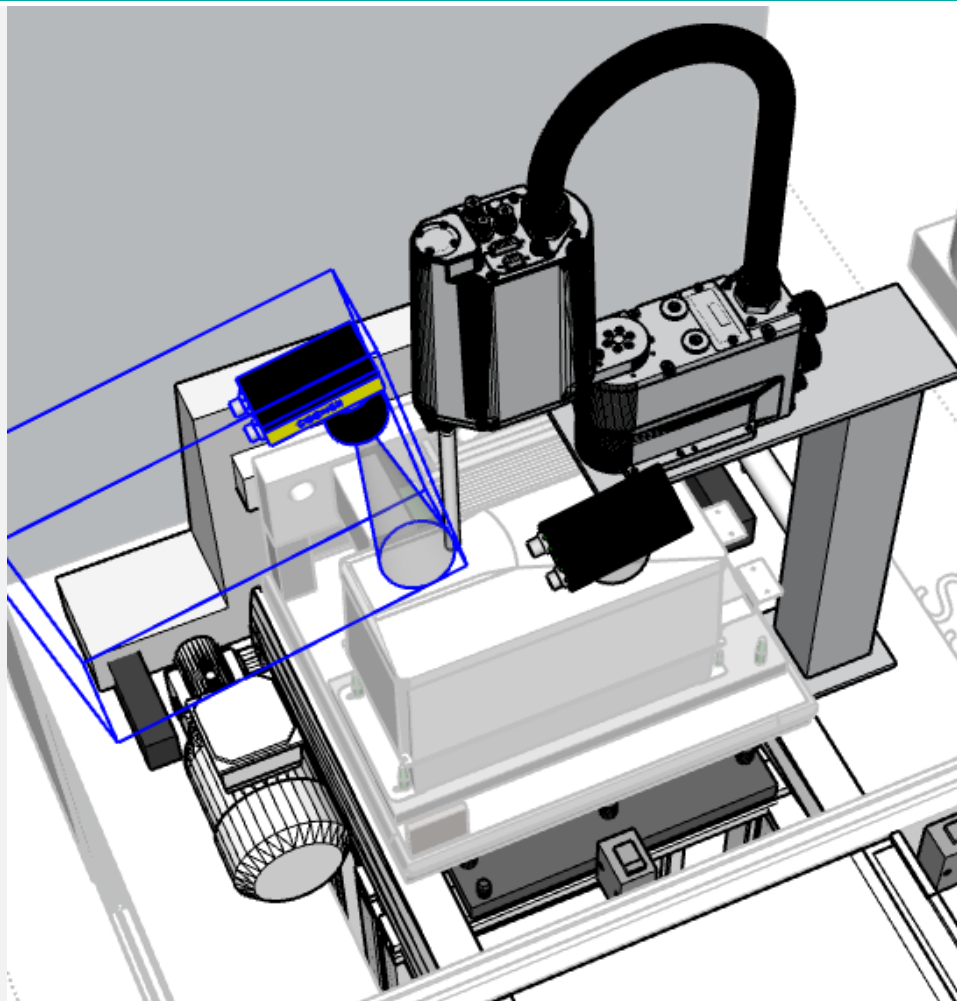
### F03 : RFID code reading

#### Function details

- 1 x Supporting frame
- 1 x RFID code reading unit

**F04 : Drive pushbuttons external automated actuation**

**Function details**

- 1 x Supporting frame
- 1 x DENSO 4 axis robot (reach > 200mm) controller, cables (or similar brand)  
*Ensuring flexibility for several drive models and HMI typologies.*
- 1 x Push tool with soft terminal ensuring no damage on HMI buttons (pushing on HMI buttons)
- 1 x Remote I/O

**F05 : Drive HMI vision check****Function details**

1 x Supporting frame

1 x Camera, wiring, optics, lightning

*Camera field of view will include drive HMI and strictly surrounding area.*

**F06 : Additional vision check****Function details**

1 x Supporting frame

1 x Camera, wiring, optics, lightning

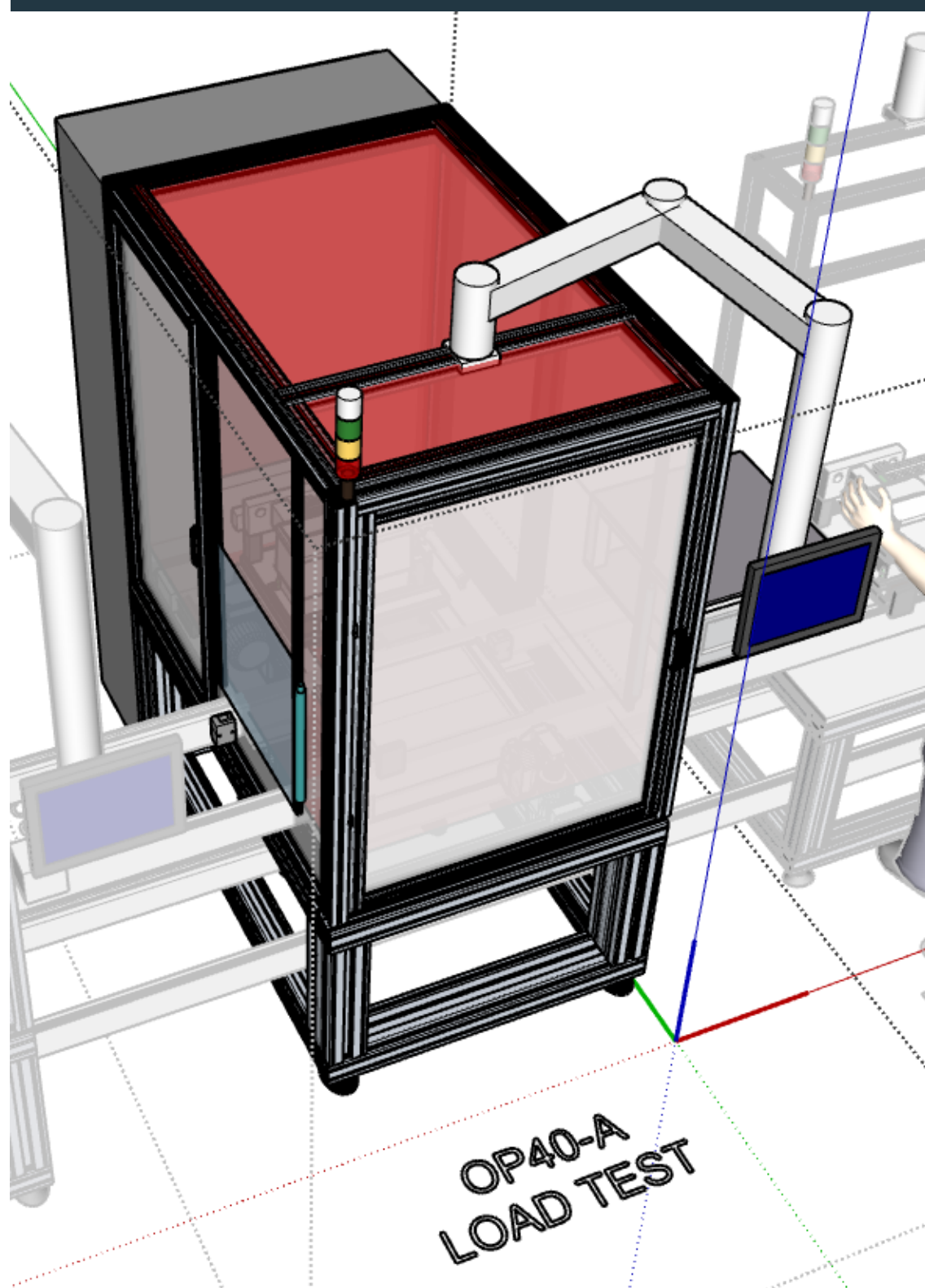
*Subject of vision test and field of view to be implemented TBD.*



**F07 : Electrical test**
**Function details**

- 1 x Movement system for connection unit (pneumatic)
- 1 x Connection unit towards pallet match plate (STAUBLI Combitaq or similar brand/model).
- 1 x Industrial PC with I/O boards
- 1 x Siemens SINAMICS S120 Grid Simulator system (or similar brand/model) configured to deliver 220V, 440V, and 575V AC outputs up to 60A from the 380V/60Hz supply  
*Includes a first AC/DC conversion stage, an ALM as DC/AC conversion stage, an Active Interface Module (AIM) sinusoidal filter. This preliminary hardware configuration represents our baseline technical proposal to meet the highlighted power requirements. The exact scope of supply and component sizing are to be confirmed or varied (even simplified) based on the final detailed specifications.*
- 1 x We consider that electronic load to be used in this station we'll be delivered by WEG.  
*We include interface and coupling hardware to integrate the electronic loads supplied by WEG automation.*
- 1 x Acquisition of voltage and current on 3 x AC IN channels + 3 x AC OUT channels. About current measurements we'll use LEM probes or similar.  
*This item has been defined regarding the specification requirement: motor operation and output current measurement.*  
*Here we are not considering use of power analyzer.*  
*Basing on detailed specification receipt we can confirm or modify this proposal item.*
- 3 x Communication Bus Test Gateways  
*Multiplexer interface system for Modbus RS485, Ethernet port, USB communication validation.*  
*To be analysed need for eventual additional Bus typologies here not considered.*
- 1 x Resistor to test the drive's internal chopper/BR terminal.  
*Equipment to be defined according detailed specification.*

- 1 x STO (Safe Torque Off) Verification Module  
*Automated relay circuit and diagnostic instrumentation to validate the drive's safety stop module.*
- 1 x Test of ground fault and output short-circuit protections  
*Automated relay circuit to validate what in object.*
- 1 x Remote I/O

**OP40A : Load test module**

**OP40A : General equipment**
**F01 : Machine structure**
**Function details**

- 1 x Aluminium profiles bottom structure featured by dimensions showed on layout
- 1 x Upper structure realized with aluminium profiles and **thermal insulation panels**. Designed in consdieration of **max temperature = 50°C**.
- 1 x Steel base plate
- 1 x Pneumatic protection panel (right side); open during pallet movement / closed during testing. **It includes thermal insulation panel.**
- 1 x Pneumatic protection panel (left side); open during pallet movement / closed during testing. **It includes thermal insulation panel.**
- 3 x Door interlock
- 1 x Pushbuttons, emergency buttons and other general electrical accessories
- 1 x Machine lighting
- 1 x Pushbutons, Lightning, Andon light (showing machine status: emergency or failure / automatic mode / piece waiting)
- 1 x Air treatment unit, flow switch, shut-off valve enabling air interception upstream machine.

**F02 : Operator panel**
**Function details**

- 1 x Supporting arm
- 1 x WEG 15,6"" cMTX series HMI

**F03 : Electrical cabinet**
**Function details**

- 1 x Cabinet, general sectioning, safety and protection components, indicator lights, user sockets. Cabinet power supply will be: 380 Vac 60 Hz power supply (auxiliaries: 24 VDC), PLC WEG. The PLC will perform essentially the pallet IN/OUT and start cycle. The test routine will be fully executed by PC.

**OP40A : Functional equipment**
**F01 : Pallet conveying into station**
**Function details**

- 1 x TS2 Bosch Rexroth 90° transverse conveyor unit (or similar brand/model)  
+ 700mm reversible conveyor segment
- 1 x Remote I/O, motors management, accessories
- 1 x Pneumatic valves and accessories

**F02 : Pallet management**
**Function details**

- 3 x Pallet stop pneumatic device
- 1 x Pneumatic index device for pallet in test position
- 1 x Remote I/O
- 1 x Pneumatic valves and accessories

**F03 : RFID code reading**
**Function details**

- 1 x Supporting frame
- 1 x RFID code reading unit

**F04 : Module temperature regulation system**
**Function details**

- 1 x Forced convection heating unit with PID controller and sensor to maintain cabin at 50°C.
- 1 x Remote I/O

**F05 : Electrical test**
**Function details**

- 1 x Movement system for connection unit (pneumatic)
- 1 x Connection unit towards pallet match plate (STAUBLI Combitaq or similar brand/model).
- 1 x Industrial PC with I/O boards

- 1 x Siemens SINAMICS S120 Grid Simulator system (or similar brand/model) configured to deliver 220V, 440V, and 575V AC outputs up to 60A from the 380V/60Hz supply  
*Includes a first AC/DC conversion stage, an ALM as DC/AC conversion stage, an Active Interface Module (AIM) sinusoidal filter. This preliminary hardware configuration represents our baseline technical proposal to meet the highlighted power requirements. The exact scope of supply and component sizing are to be confirmed or varied (even simplified) based on the final detailed specifications.*
- 1 x We consider that electronic load to be used in this station we'll be delivered by WEG.  
*We include interface and coupling hardware to integrate the electronic loads supplied by WEG automation.*
- 1 x Yokogawa WT5000 Power analyzer (or similar brand/model), 7x 760902 IN modules, 7 x LEM or similar brand current probes, interface wiring and accessories.  
*Instrumentation for realtime acquisition of voltage and current related with: 3 x AC-IN (L1, L2, L3); 3 x AC-OUT (U,V,W): 1 x CC.*  
*The exact scope of supply and component sizing are to be confirmed or varied (even simplified) based on the final detailed specifications.*
- 1 x Remote I/O
- 1 x Pneumatic management

**S : Server**

Solution for centralized management and monitoring of testing data. The system structure is composed of a **SQL Server** - Relational database management system (DBMS) for efficiently storing, organizing, and accessing data.

The Server provides functionalities to automate data collection and analysis, facilitating the management of the following activities through dedicated interfaces:

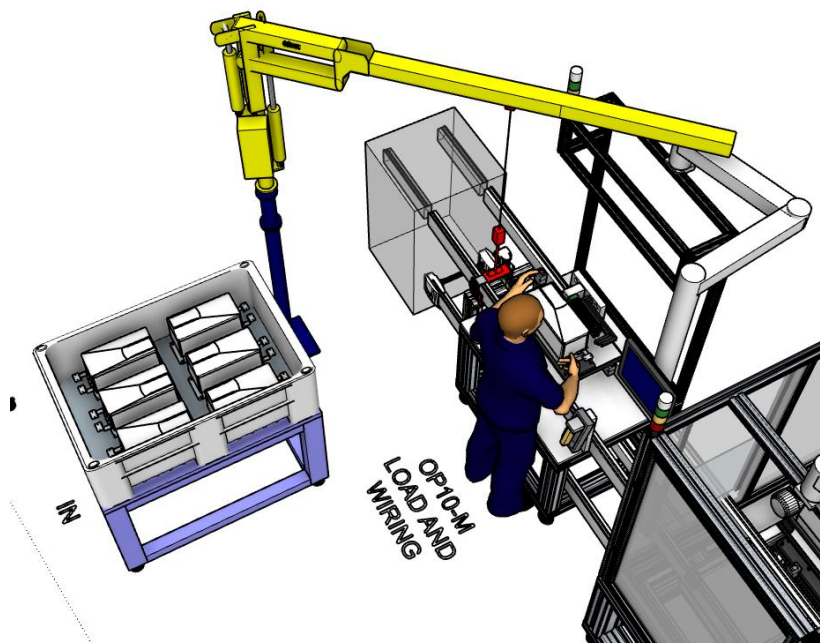
- BOM (Bill of Material) management
- Configuration and management of stations parameters
- Data collection and storage on SQL Server
- Data transmission via DataRelay

Server will be designed to interface with external company systems, allowing data sharing based on customer needs.



## OPTIONS

### Opt.1 - OP10M : Load and wiring module



### OP10M : General equipment

#### F01 : Machine structure

##### Function details

- 1 x Aluminium profiles bottom structure featured by dimensions showed on layout
- 1 x Upper structure (manual station) realized with aluminium profiles
- 1 x Pushbuttons, emergency buttons and other general electrical accessories
- 1 x Machine lighting
- 1 x Pushbuttons, Lightning, Andon light (showing machine status: emergency or failure / automatic mode / piece waiting)

#### F02 : Operator panel

##### Function details

- 1 x Supporting arm
- 1 x WEG 15,6" cMTX series HMI

**F03 : General electrical equipment**
**Function details**

1 x Remote I/O (not included at function level)

**F04 : Pneumatic valves and relevant accessories**
**Function details**

1 x Pneumatic valves and relevant accessories

**OP10M : Functional equipment**
**F01 : Pallet management**
**Function details**

2 x Pallet stop pneumatic device

1 x Remote I/O

1 x Pneumatic valves and accessories

**F02 : RFID code reading**
**Function details**

1 x Supporting frame

1 x RFID code reading unit

**F03 : 2D Code reading**
**Function details**

1 x Supporting frame

1 x Handheld 2D code reader

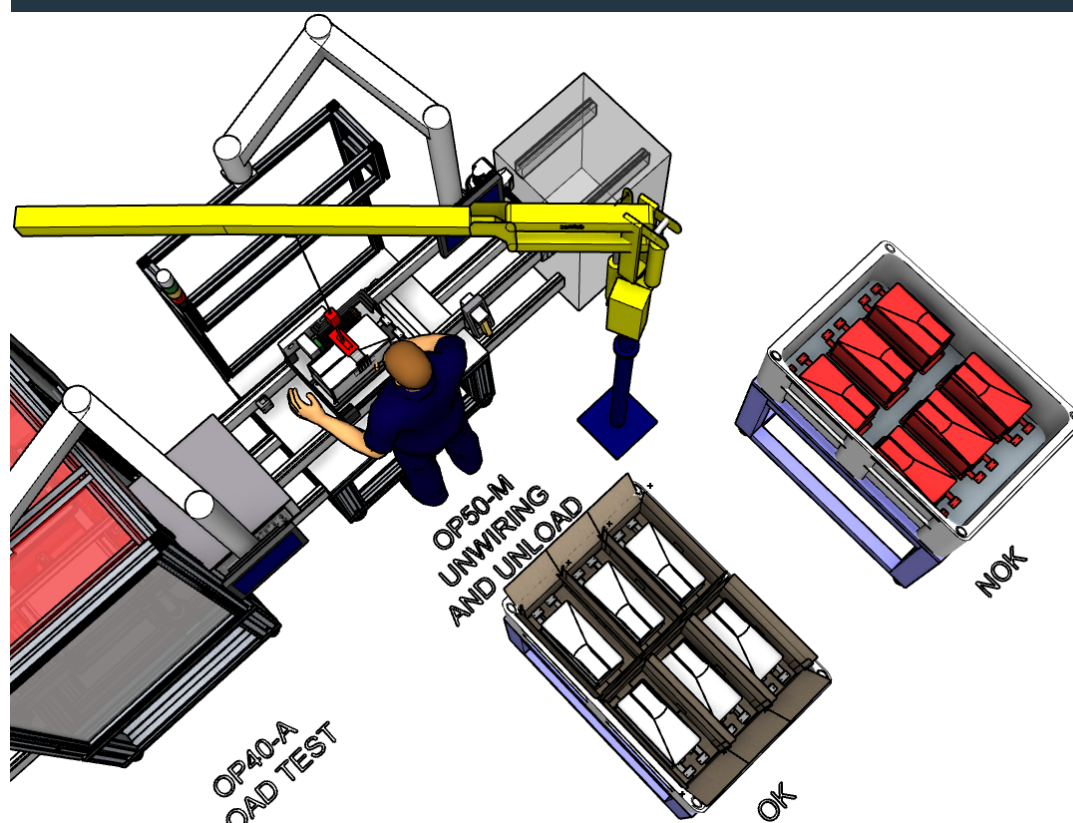
**F04 : Manual handling system**
**Function details**

1 x DEMAG (or similar brand) chain manipulator. Manual gripping function TBD.

1 x Remote I/O

1 x Pneumatic valves and accessories

## OP50A : Unwiring and unload module



## OP50A : General equipment

### F01 : Machine structure

#### Function details

- 1 x Aluminium profiles bottom structure featured by dimensions showed on layout
- 1 x Upper structure (manual station) realized with aluminium profiles
- 1 x Pushbuttons, emergency buttons and other general electrical accessories
- 1 x Machine lighting
- 1 x Pushbuttons, Lightning, Andon light (showing machine status: emergency or failure / automatic mode / piece waiting)

### F02 : Operator panel

#### Function details

- 1 x Supporting arm
- 1 x WEG 15,6" cMTX series HMI

**F03 : General electrical equipment**
**Function details**

1 x Remote I/O (not included at function level)

**F04 : Pneumatic valves and relevant accessories**
**Function details**

1 x Pneumatic valves and relevant accessories

**OP50A : Functional equipment**
**F01 : Pallet management**
**Function details**

2 x Pallet stop pneumatic device

1 x Remote I/O

1 x Pneumatic valves and accessories

**F02 : RFID code reading**
**Function details**

1 x Supporting frame

1 x RFID code reading unit

**F03 : 2D Code reading**
**Function details**

1 x Supporting frame

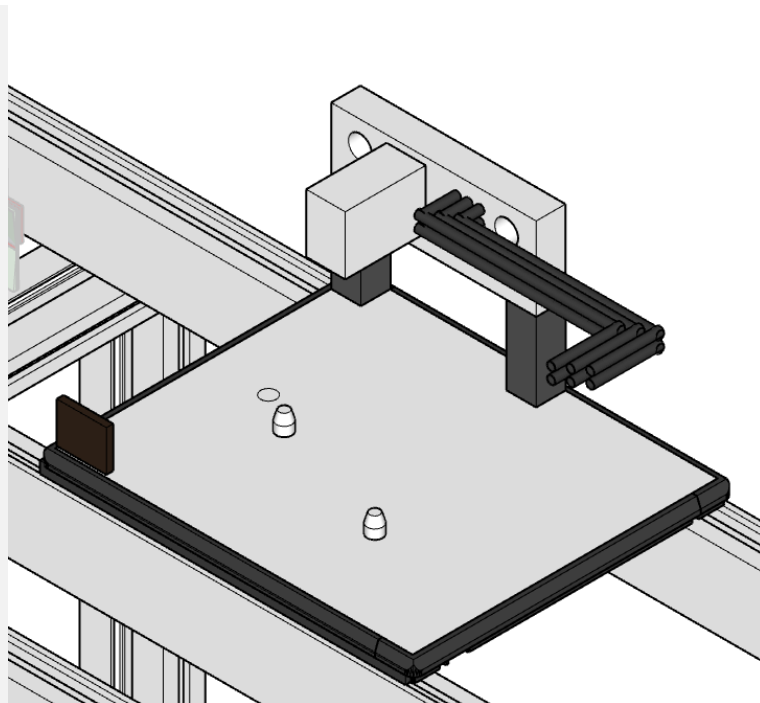
1 x Handheld 2D code reader

**F04 : Manual handling system**
**Function details**

1 x DEMAG (or similar brand) chain manipulator. Manual gripping function TBD.

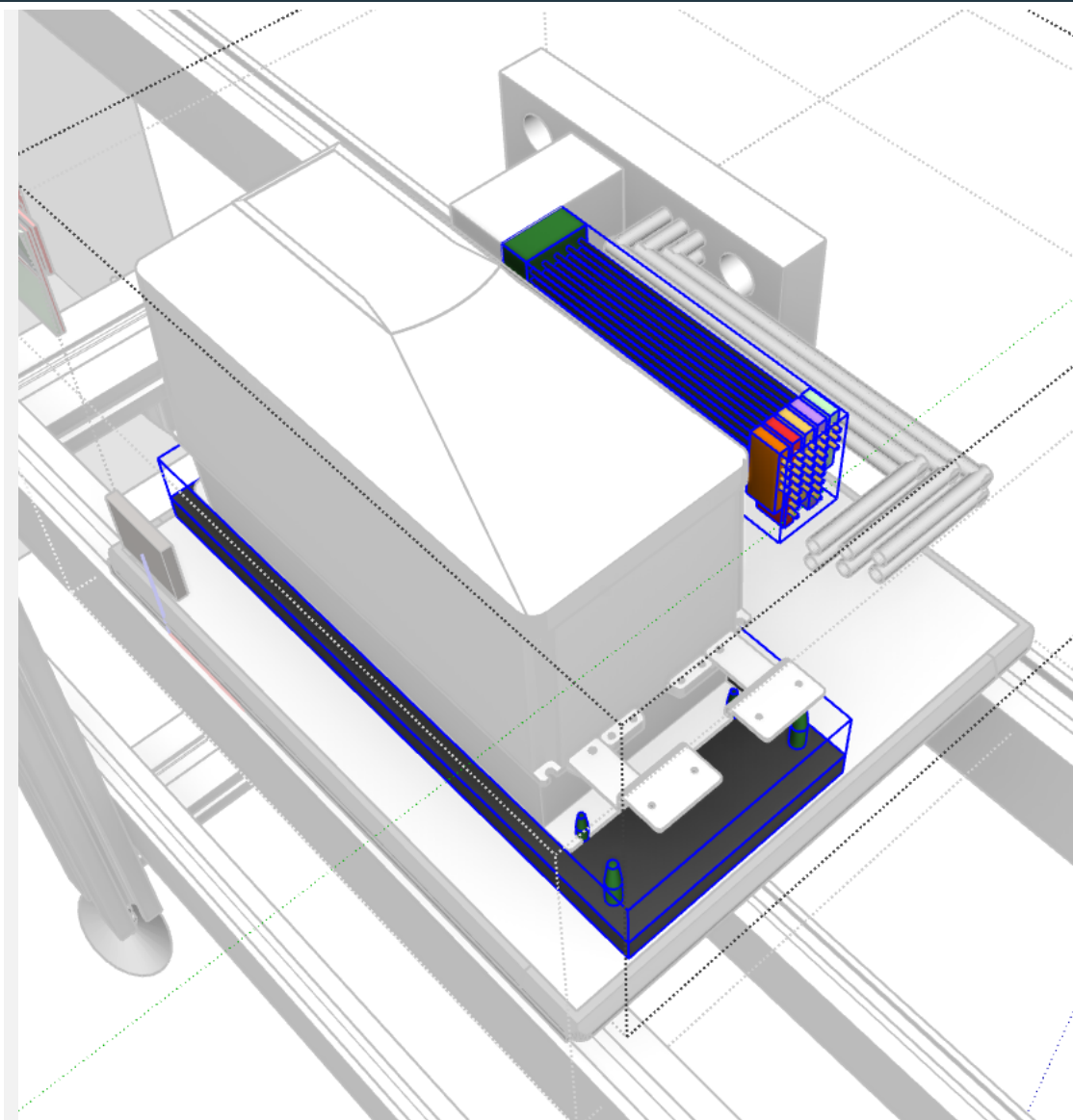
1 x Remote I/O

1 x Pneumatic valves and accessories

**00 : Additional pallet****F01 : Series pallet****Function details**

- 1 x Pallet base featured by dimensions  $\leq 400 \text{ mm} \times 480 \text{ mm}$
- 1 x Base pallet machining and centring pins (2x) for changeover fixture nests
- 1 x Match plate (STAUBLI Combitaq or similar brand/model) including 3 x AC IN connections + 3 x AC OUT connections + signal connections (D-I/O, A-I/O, Bus, STO etc) & AC HV harness & LV connector for changeover harness
- 1 x RFID TAG memory

**00 :      Changeover equipment: for 1 typology, for 1 pallet - rough  
BUDGET price**



#### **Function details**

- 1 x    Changeover fixture nest including centring pins for specific drive model.  
Fixtures will include 1 QR code for typology check at loading station.
- 1 x    Changeover harness for signal connections (D-I/O, A-I/O, Bus, STO etc)

## IMPORTANT DETAILS

- We offer the opportunity to conduct comprehensive design review and validation sessions on-site at Masmec, leveraging advanced **Virtual Reality (VR) tools to streamline evaluation and foster multidisciplinary collaboration.**
- Option 1 (stations OP10-M / OP50-M) do not include equipment for harness assembly/disassembly (screwdrivers etc.); if asked we can implement it with extra price.
- Once drive QR code has been read on OP10-M, relevant information will be written on pallet TAG; this will avoid to implement additional QR code readers on test stations.
- If will not be purchased Option 1 item for drive manual load/unload (that includes also manual QR reading) we'll have to implement additional fix QR reading within base proposal (upstream HIPOT test module).
- For each test station we consider to implement n.1 RFID read/write device on the pallet position where pallet can move IN/OUT with respect to conveyor main flow.

## IMPORTANT TOPICS IN ORDER TO MOVE FROM BUDGET QUOTE TO FORMAL QUOTE

- If we move towards a detailed quote, it is important to get information about each single drive typology to be tested, versions to be tested, communication BUS typologies to be tested, connectors (according base drive versions and eventual drive option items to be tested) etc.
- Need of 2D drawings of features to be interfaced by our test equipment (ex. supporting features / HV&LV connectors / HMI /) besides overall 2D drawing of all drive models to be processes.
- In general we need the detailed test specification for each test module (Hipot / Functional / Load) in order to confirm or modify this budget proposal.
- For each test station, a detailed connection schema has to be defined according the specific test sequence to be implemented. After this schema definition we can cross-check offer budget now shared.

- With reference to the sentence: "The module shall use electronic loads as load for the drives, which shall be supplied by WAU", we need of indications related with HW/SW integration of the electronic load in order to confirm or modify our proposal.
- We consider that electronic load for Load test station we'll be delivered by WEG.
- Considering the drives diversities and requirements for test in power conditions also for the Functional test station, we consider WEG availability to deliver an electronic load also with regard to this station.
- About programming functions of Functional test module we need detailed specification for HW/SW to be implemented.
- We consider that master parts will be produced through series parts and will be delivered by WEG.
- Q.ty of master drives OK / NOK to be defined.
- Q.ty of benches to be defined on the basis of the aimed production capacity.
- Eventual import TAXs are not included in this quote.
- Considering that main conveyor has been already quoted as OP00, the stations (OP10,20,30,40,50) do not include a main flow conveyor segment.
- Detailed spares list will be defined at end of design phase. If necessary, an option with budget for spare parts can be added to the offer.



## REMOTE MONITORING AND ASSISTANCE

The offer includes a free remote assistance service during the warranty period, during MASMEC working hours (8:00-17:00 CET, Monday to Friday). This support allows software interventions on the machine directly from our headquarter. Upon request, it is possible to organize technical assistance outside working hours, even during the warranty period, with methods and costs to be agreed.

### Configuring Remote Assistance Services

To activate remote assistance, the customer must have a company IP address with internet access and the necessary hardware and software.

## REJECTED PIECES MANAGEMENT

If a part will be rejected, the operator in OP50-M (option station), will take the piece from the holding fixture and place it in the reject box.

## FAILURE MANAGEMENT

Failure conditions will be explained with a message on the HMI about the point of the machine where the failure occurred.

## DATA STORAGE

By means of our software it is possible to carry out time referred analysis of the results of the tests made on the machines, also allowing the solution of the problem of the long term saving of the historic archives.

The software is capable to export test data towards other software platform as WINDOWS/EXCEL, in ASCII format.

It will be possible to display on the monitor and print all the data.

In detail the following statistical functions will be provided:

- test history;
- tests report;
- archives maintenance.

## PRODUCTIVITY

Line cycle time: TBD on the basis of modules q.ty.

MASMEC commits below standard values:

1. To guarantee Line Quality rate  $\geq 99\%$ , calculated as the mean value of each station Quality rate after excluding rejected pieces due to faulty products or operator error.
2. To guarantee a Technical availability of each single test station  $\geq 95\%$  after excluding inefficiencies which MASMEC is not responsible for.

Terms above:

- have to be intended **due to losses caused by Masmec equipment**
- can be ensured only with parts perfectly in conformity with drawings and without defects (debris, deformations, etc.)

## MATERIALS

We will use the following materials:

- steel burnished tooling, heat treated in the parts more subject to wear;
- polycarbonate panels safety guards;
- FESTO pneumatic cylinders;
- WEG PLC;
- WEG operating panels;

With regard to the components not mentioned in the previous list, the most appropriate choices will be made together.

You will have to provide the following components, **with transport at Your charge to our plant in Bari:**

- part samples for tests and set up;
- fluid for tests (when eventually necessary).

The costs for the eventual disposal of materials no longer necessary will be communicated and charged to you if you will not decide to collect the mentioned materials.

## REGULATIONS

Making reference to the parts MASMEC will supply and/or modify, the process will respond to the safety criteria provided by the applicable European directives and, in detail, to the following:

- Directive 2006/42/EC of the European Parliament and Council of 17 May 2006 on machinery (MD – Machinery Directive);

MASMEC will supply a written CE declaration (Type A) (ref. Machine Directive 2006/42/EC, Annex II) at the end of the work. The machine will be CE certified.

**We make the point that** the elements we will supply will be designed and manufactured in order not to be source to the outside of potentially explosive atmosphere and that will have to be located in areas classified as “UNCLASSIFIED ZONE” according to the current European directive 2014/34/EU only.

## TRAINING

During the start-up of the equipment at your plant a training session will be executed.

The session will be on the following topics:

- basic information about the systems supplied;
- different ways of operating;
- most frequent regulations;
- safety procedures;
- preventive maintenance;
- wiring diagram of the control panel, power circuit and control equipment.

## DOCUMENTATION

30 days after the acceptance at Your plant, MASMEC will send the following documentation on **electronic format**:

- 3D Model of each station (STEP/PARASOLID file);
- 2D drawings of mechanical functions and parts (PDF file); 2D overall drawings of stations and/or modules are available with extra-price;
- mechanical parts subject to wear (excel file);
- electric and pneumatic schemes, comprehensive of part list in English language (PDF file);
- software list;
- commercial software license;
- user instructions (**supplied also in no.1 hard copy**);

MASMEC will provide N.1 Maintenance and instruction manual in paper form in Italian language and N.1 Maintenance and instruction manual in paper form in User official language (for EU countries) and in English language (for countries outside the EU). MASMEC could quote with extra-price the translation of Maintenance and instruction manual in different language than the official language according to regulations.

All the drawings will be developed:

- according to European drawing regulation
- with MASMEC title block
- with MASMEC numeration criteria
- with English/Italian language

As option we could quote with extra-price the drawings development with your title block, your numeration criteria and different language.

Commercial components documentation will be in the manufacturer official language and/or in the languages you will ask, as far as possible. We will not translate commercial components documentation.

In order to reduce environment impact, MASMEC will minimize use of paper. If you desire, we could quote with extra-price the requested documentation in paper form. Other specific documents, not explicitly mentioned, will be quoted with extra-price.

## SUPPLIES

Here below follows a summary of needed supplies:

|                         |                                  |
|-------------------------|----------------------------------|
| <b>Control system</b>   | 380 Vac, 3ph, 60 Hz power supply |
| <b>Pneumatic system</b> | 5,5 bar air supply               |

## WARRANTY TERMS

### Obligations of Masmec

In accordance with terms described in this paragraph, Masmec will provide machines or parts in compliance with technical specifications and agreements and without any defect which would make them unsuitable for their intended use.

The warranty covers material flaws, assembly flaws or construction flaws.

### Warranty duration

The warranty period on the equipment supplied by MASMEC is of 12 months from machine reception and, in any case, it will not exceed 14 months from delivery.

**If You're interested, we can quote separately with extra-price a 12 months extension of warranty coverage period.** The warranty extension covers exclusively defects generated by design, assembly and programming errors, while commercial components will be guaranteed exclusively according to the warranty terms established by the original manufacturer at the time of supply to Masmec.

### Restrictions

Warranty does not include:

- lamps, fuses, belts, filters, sealing joints, retainers, lubricants, batteries, work lights, reactors, glass, push buttons;
- consumables and material subject to wear;
- ordinary maintenance operations;
- travelling, lodging and boarding expenses of our technicians;
- coverage of lost production, scrap, etc.;
- defects caused by non-compliance with instruction and maintenance manual or, in general, by negligence or incorrect use of equipment;

- defects originated in case of equipment not used in the normal conditions stipulated in the specification or the commercial proposal;
- anomalies or malfunctioning due to errors made during machine installation or part mounting, if installation/mounting is not performed directly by Masmec or however under the supervision of our specialized staff;
- defects or malfunctions of existing machines for which retooling or revamping activities have been carried out. For such machines, the warranty will be limited exclusively to the new parts installed by Masmec;
- damages due to inadequate warehousing;
- materials provided to MASMEC by the Customer;
- problems arising from an act of God or unforeseen circumstance.

Technical assistance will be completely at Your charge, when guarantee period has expired.

**Non-compliance issues notification**

Customer will have to notify to Masmec the non-compliance issue specifying the problem in a written minute within 15 days from the date of detection and It will allow to perform all the required checks; otherwise guarantee cover could be lost. Written notification of a flaw shall not exempt the Customer from his contractual obligations and from meeting the agreed terms of payment.

**Reparations or substitutions**

After receiving Your regular notification about issue and proceeding with related verification, Masmec will:

- a) supply to You the needed parts free of charge in order to substitute the malfunctioning ones, or
- b) execute directly the reparation free of charge or delegate it to a third subject at our charge.

**PRICES (OUR BUDGET QUOTE C610222-1)**

The prices, VAT excluded, for study and supply the above mentioned equipment are the following:

**M.1 - Base proposal:**

| <b>POS.</b>                                                                                    | <b>DESCRIPTION</b>                                                                                                                                               | <b>BUDGET PRICES</b> |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>OP00</b>                                                                                    | <b>Conveyor system, pallets (3) and relevant changeover equipment, masters management</b><br><i>Conveyor length as detailed in proposal (9m top +9m bottom).</i> | <b>€ 270.000</b>     |
| <b>OP20A</b>                                                                                   | <b>HIPOT test module</b>                                                                                                                                         | <b>€ 190.000</b>     |
| <b>OP30A</b>                                                                                   | <b>Functional test module</b>                                                                                                                                    | <b>€ 370.000</b>     |
| <b>OP40A</b>                                                                                   | <b>Load test module</b>                                                                                                                                          | <b>€ 380.000</b>     |
| <b>S</b>                                                                                       | <b>Server</b>                                                                                                                                                    | <b>€ 40.000</b>      |
| <b>TOT. BUDGET PRICE</b><br><i>Installation<br/>at Santa Catarina plant</i><br><b>INCLUDED</b> |                                                                                                                                                                  | <b>€ 1.250.000</b>   |

- Prices above valid for items simultaneous purchase at first base order
- 5% discount to be applied to above modules prices for each additional module purchased within 2027

**Options:**

| <b>POS.</b>   | <b>DESCRIPTION</b>                                                                                                    | <b>BUDGET PRICES</b> |
|---------------|-----------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Opt.1</b>  | <b>Manual wiring and load/unload (OP10-M, OP50-M)</b><br><i>Price valid for purchase contextual with base order.</i>  | <b>€ 150.000</b>     |
| <b>Opt.2A</b> | <b>1 week on-site support (1 technician x 5 working days on central shift, travel and boarding expenses included)</b> | <b>€ 9.500</b>       |

|               |                                                                                                                         |                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Opt.2B</b> | <b>1 week extension of on-site support (1 technician x 5 working days on central shift, boarding expenses included)</b> | <b>€ 6.500</b>    |
| <b>Opt.3A</b> | <b>Additional pallet –<u>BUDGET</u> price</b>                                                                           | <b>€ 6.000,00</b> |
| <b>Opt.3B</b> | <b>Changeover equipment: for 1 typology, for 1 pallet –<u>BUDGET</u> price</b>                                          | <b>€ 4.000</b>    |

## SUPPLY CONDITIONS

The price of the above-mentioned line includes the design, assembly, installation and start up assistance at your plant at Santa Catarina (Brazil).

Installation and training will be executed in maximum n°15 working days. If these activities should lengthen over n°15 days, for reasons not due to us, the extra-days will be quoted at final balance according to current UCIMU's tariffs.

The installation will be performed during weekdays and in central shift only (h. 8.00 – 17.00 with n°1 hour break). We are available to discuss with You other Your possible needs.

**Our supply does not include anything not explicitly mentioned in this proposal.**

|                         |                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-acceptance</b>   | At our plant.                                                                                                                                                                                                                                                                                                                                                         |
| <b>Installation</b>     | Excludes: <ul style="list-style-type: none"> <li>• unloading from means of transport;</li> <li>• any necessary hiring (fork lift trucks, transpallets, etc.);</li> <li>• positioning the machine in the required area;</li> <li>• connecting to the electric, pneumatic, water, etc. mains.</li> <li>• suction, discharge, etc.;</li> <li>• building work.</li> </ul> |
| <b>Final acceptance</b> | At Your plant.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Warranty</b>         | See "Warranty terms" paragraph.                                                                                                                                                                                                                                                                                                                                       |
| <b>Shipment</b>         | <b>50 weeks ARO</b>                                                                                                                                                                                                                                                                                                                                                   |



|                             |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical assistance</b> | <p>Within 48 hrs during weekdays, rectified quotes according to UCIMU prices in force (<b>for European Countries</b>).</p> <p>Within 72 hrs during weekdays, bound up with Visa released by the foreign embassy (where required), rectified quotes according to UCIMU prices in force (<b>for other Countries</b>).</p>                                                                                                   |
| <b>Transport</b>            | <b>FCA</b> Bari - Italy (Incoterms® 2020)                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Packing</b>              | Included, MASMEC standard for sea freight                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Payment</b>              | <p>to be agreed</p> <p><b>MASMEC highlights that if validation process is not completed within 30 days of planned date, for reasons not imputable to MASMEC (e.g production scheduling, Components Non-conformity / Unavailability, Operators Availability, other reasons, etc.), an “acceptance report with reserve” needs to be signed by You. In this way, the corresponding payment tranche will be released.</b></p> |
| <b>Validity</b>             | 30 days                                                                                                                                                                                                                                                                                                                                                                                                                   |

Please find attached hereafter the proposed planning for the scheduled activities of this project.

**PLANNING**

|    | Week from order                              | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |  |  |
|----|----------------------------------------------|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|--|
| 1  | Order                                        |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 2  | Receipt of final drawings and technical spec |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 3  | Kick off meeting                             |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 4  | Design                                       |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 5  | Design review                                |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 6  | Design validation                            |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 7  | Software design offline                      |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 8  | Purchasing                                   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 9  | Procurement                                  |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 10 | Samples reception @ MASMEC                   |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 11 | Equipment at customer charge @ MASMEC        |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 12 | Equipment assembly                           |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 13 | Software online                              |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 14 | Tests                                        |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 15 | Validation at Masmec                         |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 16 | Final Tuning after validation                |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |
| 17 | Shipment                                     |   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |  |

Best regards

**MASMEC**

**MASMEC SPA**

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